

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673



231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

March 24, 2026

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 24H0470
Client Project NPDES AMR.- August 2024

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

A handwritten signature in black ink that reads "Timothy W. Brewer".

Tim Brewer
Project Manager
Enclosures



TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
24H0470-01	TERT. Effluent (001A)	Wastewater	DS/TL	08/08/24 07:00	08/08/24 13:50
Sample type: Composite					
24H0470-02	TERT. Effluent (001A)	Wastewater	DS/TL	08/08/24 07:00	08/08/24 13:50
Sample type: Composite					
24H0470-03	TERT. Effluent (001A)	Wastewater	DS/TL	08/08/24 07:00	08/08/24 13:50
Sample type: Composite					
24H0470-04	TERT. Effluent (001A)	Wastewater	DS/TL	08/08/24 10:22	08/08/24 13:50
Sample type: Grab					
24H0470-05	TERT. Effluent (001A)	Wastewater	DS/TL	08/08/24 10:20	08/08/24 13:50
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-01 Matrix: Wastewater Date Collected: 08/08/24 07:00
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.7 Rev. 4.4									
Batch: T154824									
Calcium	72 mg/L	1.0	1	08/12/24	fs	08/15/24	acs		
Magnesium	19 mg/L	1.0	1	08/12/24	fs	08/15/24	acs		
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T154824									
Antimony	0.66 ug/L	1.0	1	08/12/24	fs	08/13/24	jlh		
Chromium	0.63 ug/L	5.0	1	08/12/24	fs	08/13/24	jlh	J	
Lead	<1.0 ug/L	1.0	1	08/12/24	fs	08/13/24	jlh		
Nickel	2.9 ug/L	5.0	1	08/12/24	fs	08/13/24	jlh	J	
Selenium	0.86 ug/L	1.0	1	08/12/24	fs	08/13/24	jlh	J	
Thallium	<1.0 ug/L	1.0	1	08/12/24	fs	08/13/24	jlh		
Zinc	13 ug/L	10	1	08/12/24	fs	08/13/24	jlh		
Analysis Method: SM 2340 B-21									
Batch: [CALC]									
Hardness as CaCO3	260 mg/L	2.6	1	08/12/24		08/15/24	acs	N	

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-02 Matrix: Wastewater Date Collected: 08/08/24 07:00
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
<i>Batch: T154793</i>									
N-Nitrosodimethylamine	<4.8 ug/L	4.8	1	08/09/24	kbc	08/12/24	drm		
Bis(2-chloroethyl)ether	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
2-Chlorophenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Phenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
1,4-Dichlorobenzene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Bis(2-chloroisopropyl)ether	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
N-Nitrosodi-n-propylamine	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Nitrobenzene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Isophorone	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2-Nitrophenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2,4-Dimethylphenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Bis(2-chloroethoxy)methane	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
1,2,4-Trichlorobenzene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2,4-Dichlorophenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Naphthalene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
4-Chloro-3-methylphenol	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2,4,6-Trichlorophenol	<1.7 ug/L	1.7	1	08/09/24	kbc	08/12/24	drm		
2-Chloronaphthalene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Dimethyl phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Acenaphthylene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2,6-Dinitrotoluene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Acenaphthene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
2,4-Dinitrotoluene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
4-Nitrophenol	<19 ug/L	19	1	08/09/24	kbc	08/12/24	drm		
2,4-Dinitrophenol	<19 ug/L	19	1	08/09/24	kbc	08/12/24	drm		
Diethyl phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Fluorene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
4-Chlorophenyl phenyl ether	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
4,6-Dinitro-2-methylphenol	<4.8 ug/L	4.8	1	08/09/24	kbc	08/12/24	drm		
N-Nitrosodiphenylamine	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
1,2-Diphenylhydrazine	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm	N	
4-Bromophenyl phenyl ether	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-02 Matrix: Wastewater Date Collected: 08/08/24 07:00
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Phenanthrene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Anthracene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Di-n-butyl phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Fluoranthene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Benzidine	<48 ug/L	48	1	08/09/24	kbc	08/12/24	drm		
Pyrene	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Butyl benzyl phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Benzo (a) anthracene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Chrysene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Bis(2-ethylhexyl)phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Di-n-octyl phthalate	<2.4 ug/L	2.4	1	08/09/24	kbc	08/12/24	drm		
Benzo (b) fluoranthene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Benzo (k) fluoranthene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Benzo (a) pyrene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Indeno (1,2,3-cd) pyrene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Dibenz (a,h) anthracene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Benzo (g,h,i) perylene	<0.97 ug/L	0.97	1	08/09/24	kbc	08/12/24	drm		
Surrogates:									
2-Fluorophenol	27 %	20-53	1	08/09/24	kbc	08/12/24	drm		
Phenol-d5	14 %	11-40	1	08/09/24	kbc	08/12/24	drm		
Nitrobenzene-d5	58 %	36-103	1	08/09/24	kbc	08/12/24	drm		
2-Fluorobiphenyl	60 %	36-119	1	08/09/24	kbc	08/12/24	drm	N	
2,4,6-Tribromophenol	65 %	30-105	1	08/09/24	kbc	08/12/24	drm	N	
Terphenyl-d14	59 %	37-109	1	08/09/24	kbc	08/12/24	drm		
Analysis Method: EPA 625.1 SIM									
<i>Batch: T154928</i>									
Benzidine	<0.099 ug/L	0.099	1	08/13/24	jh	08/14/24	drm		
3,3'-Dichlorobenzidine	<0.99 ug/L	0.99	1	08/13/24	jh	08/14/24	drm		
Surrogates:									
Nitrobenzene-d5	82 %	38-130	1	08/13/24	jh	08/14/24	drm		
2-Fluorobiphenyl	95 %	39-118	1	08/13/24	jh	08/14/24	drm	N	
Terphenyl-d14	70 %	14-96	1	08/13/24	jh	08/14/24	drm		

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-03 Matrix: Wastewater Date Collected: 08/08/24 07:00
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
PESTICIDES/PCBS									
Analysis Method: EPA 612									
<i>Batch: T154856</i>									
Hexachlorobenzene	<0.0095 ug/L	0.0095	1	08/12/24	kbc	08/14/24	jh	N	
Hexachlorobutadiene	<0.0095 ug/L	0.0095	1	08/12/24	kbc	08/14/24	jh	N	
Hexachlorocyclopentadiene	0.0028 ug/L	0.0095	1	08/12/24	kbc	08/14/24	jh	J, N	
Hexachloroethane	<0.0095 ug/L	0.0095	1	08/12/24	kbc	08/14/24	jh	N	
Pentachlorophenol	<1.7 ug/L	1.7	1	08/12/24	kbc	08/14/24	jh	N	
Surrogates:									
Tetrachloro-m-xylene	71 %	18-105	1	08/12/24	kbc	08/14/24	jh	N	

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-04 Matrix: Wastewater Date Collected: 08/08/24 10:22
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
<i>Batch: T154763</i>									
Chloromethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Vinyl chloride	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Bromomethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Chloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Acrolein	<10 ug/L	10	1	08/08/24	ats	08/09/24	ats		
1,1-Dichloroethene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Methylene chloride	<5.0 ug/L	5.0	1	08/08/24	ats	08/09/24	ats		
Acrylonitrile	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
trans-1,2-Dichloroethene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,1-Dichloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Chloroform	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,1,1-Trichloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Carbon tetrachloride	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Benzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,2-Dichloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Trichloroethene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,2-Dichloropropane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
cis-1,2-Dichloroethene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Bromodichloromethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
2-Chloroethylvinyl ether	<5.0 ug/L	5.0	1	08/08/24	ats	08/09/24	ats		
cis-1,3-Dichloropropene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Toluene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
trans-1,3-Dichloropropene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,1,2-Trichloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Tetrachloroethene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Dibromochloromethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Chlorobenzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Ethylbenzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Bromoform	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,1,2,2-Tetrachloroethane	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,3-Dichlorobenzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
1,4-Dichlorobenzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-04 Matrix: Wastewater Date Collected: 08/08/24 10:22
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2-Dichlorobenzene	<1.0 ug/L	1.0	1	08/08/24	ats	08/09/24	ats		
Surrogates:									
1,2-Dichloroethane-d4	93 %	68-133	1	08/08/24	ats	08/09/24	ats		
Toluene-d8	108 %	75-120	1	08/08/24	ats	08/09/24	ats		
4-Bromofluorobenzene	98 %	69-119	1	08/08/24	ats	08/09/24	ats		
1,2-Dichlorobenzene-d4	105 %	72-127	1	08/08/24	ats	08/09/24	ats		

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ANALYTICAL RESULTS

Trace Project ID: 24H0470
Client Project ID: NPDES AMR.- August 2024

Trace ID: 24H0470-05 Matrix: Wastewater Date Collected: 08/08/24 10:20
Sample ID: TERT. Effluent (001A) Date Received: 08/08/24 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
WET CHEMISTRY									
Analysis Method: EPA 420.4 Rev. 1.0									
Batch: T155134									
Phenolics	<10 ug/L	10	1	08/15/24	pn	08/15/24	pn		

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QUALITY CONTROL RESULTS

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154824

Analysis Description: Calcium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.7 Rev. 4.4

METHOD BLANK: T154824-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Calcium	mg/L	<1.0	1.0	
Magnesium	mg/L	<1.0	1.0	

LABORATORY CONTROL SAMPLE: T154824-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Calcium	mg/L	16.00	16.8	105	85-115	
Magnesium	mg/L	16.00	17.0	106	85-115	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154824

Analysis Description: Thallium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T154824-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chromium	ug/L	<5.0	5.0	
Nickel	ug/L	<5.0	5.0	
Lead	ug/L	<0.55	0.55	
Antimony	ug/L	<0.25	0.25	
Selenium	ug/L	<0.50	0.50	
Thallium	ug/L	<0.38	0.38	
Zinc	ug/L	<10	10	

LABORATORY CONTROL SAMPLE: T154824-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chromium	ug/L	50.00	51.4	103	85-115	
Nickel	ug/L	1600	1490	93	85-115	
Lead	ug/L	100.0	103	103	85-115	
Antimony	ug/L	100.0	112	112	85-115	
Selenium	ug/L	100.0	114	114	85-115	

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LABORATORY CONTROL SAMPLE: T154824-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Thallium	ug/L	100.0	102	102	85-115	
Zinc	ug/L	1600	1570	98	85-115	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: [CALC]

Analysis Description: Hardness (Metals)

QC Batch Method:

Analysis Method: SM 2340 B-21

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154856

Analysis Description: 612 Chlorinated Hydrocarbons

QC Batch Method: EPA 3510C Separatory Funnel
Liquid-Liquid Extr.

Analysis Method: EPA 612

METHOD BLANK: T154856-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Hexachlorobenzene	ug/L	0.00650	0.010	J
Hexachlorobutadiene	ug/L	<0.010	0.010	
Hexachlorocyclopentadiene	ug/L	<0.010	0.010	
Hexachloroethane	ug/L	<0.010	0.010	
Pentachlorophenol	ug/L	<1.8	1.8	
Tetrachloro-m-xylene (S)	%	76	18-105	

LABORATORY CONTROL SAMPLE: T154856-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Hexachlorobenzene	ug/L	0.05000	0.0311	62	26-128	
Hexachlorobutadiene	ug/L	0.05000	0.0179	36	26-107	
Hexachlorocyclopentadiene	ug/L	0.05000	0.0195	39	25-105	
Hexachloroethane	ug/L	0.05000	0.0181	36	26-87	
Pentachlorophenol	ug/L	1.000	<1.8	72	25-100	J
Tetrachloro-m-xylene (S)	%	0.06000	0.0307	51	18-105	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154793

Analysis Description: TTO Semi-Volatiles

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1

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METHOD BLANK: T154793-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
N-Nitrosodimethylamine	ug/L	<5.0	5.0	
Bis(2-chloroethyl)ether	ug/L	<1.0	1.0	
2-Chlorophenol	ug/L	<2.5	2.5	
Phenol	ug/L	<2.5	2.5	
1,4-Dichlorobenzene	ug/L	<2.5	2.5	
Bis(2-chloroisopropyl)ether	ug/L	<2.5	2.5	
N-Nitrosodi-n-propylamine	ug/L	<2.5	2.5	
Nitrobenzene	ug/L	<1.0	1.0	
Isophorone	ug/L	<2.5	2.5	
2-Nitrophenol	ug/L	<2.5	2.5	
2,4-Dimethylphenol	ug/L	<2.5	2.5	
Bis(2-chloroethoxy)methane	ug/L	<2.5	2.5	
1,2,4-Trichlorobenzene	ug/L	<2.5	2.5	
2,4-Dichlorophenol	ug/L	<2.5	2.5	
Naphthalene	ug/L	<2.5	2.5	
4-Chloro-3-methylphenol	ug/L	<2.5	2.5	
2,4,6-Trichlorophenol	ug/L	<1.8	1.8	
2-Chloronaphthalene	ug/L	<2.5	2.5	
Dimethyl phthalate	ug/L	<2.5	2.5	
Acenaphthylene	ug/L	<2.5	2.5	
2,6-Dinitrotoluene	ug/L	<2.5	2.5	
Acenaphthene	ug/L	<2.5	2.5	
2,4-Dinitrotoluene	ug/L	<2.5	2.5	
4-Nitrophenol	ug/L	<20	20	
2,4-Dinitrophenol	ug/L	<19	19	
Diethyl phthalate	ug/L	<2.5	2.5	
Fluorene	ug/L	<2.5	2.5	
4-Chlorophenyl phenyl ether	ug/L	<2.5	2.5	
4,6-Dinitro-2-methylphenol	ug/L	<5.0	5.0	
N-Nitrosodiphenylamine	ug/L	<2.5	2.5	
1,2-Diphenylhydrazine	ug/L	<2.5	2.5	
4-Bromophenyl phenyl ether	ug/L	<2.5	2.5	
Phenanthrene	ug/L	<1.0	1.0	
Anthracene	ug/L	<2.5	2.5	
Di-n-butyl phthalate	ug/L	<2.5	2.5	
Fluoranthene	ug/L	<1.0	1.0	
Pyrene	ug/L	<2.5	2.5	
Butyl benzyl phthalate	ug/L	<2.5	2.5	
Benzo (a) anthracene	ug/L	<1.0	1.0	
Chrysene	ug/L	<1.0	1.0	
Bis(2-ethylhexyl)phthalate	ug/L	<2.5	2.5	
Di-n-octyl phthalate	ug/L	<2.5	2.5	
Benzo (b) fluoranthene	ug/L	<1.0	1.0	
Benzo (k) fluoranthene	ug/L	<1.0	1.0	

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METHOD BLANK: T154793-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzo (a) pyrene	ug/L	<1.0	1.0	
Indeno (1,2,3-cd) pyrene	ug/L	<1.0	1.0	
Dibenz (a,h) anthracene	ug/L	<1.0	1.0	
Benzo (g,h,i) perylene	ug/L	<1.0	1.0	
2-Fluorophenol (S)	%	28	20-53	
Phenol-d5 (S)	%	15	11-40	
Nitrobenzene-d5 (S)	%	57	36-103	
2-Fluorobiphenyl (S)	%	61	36-119	
2,4,6-Tribromophenol (S)	%	64	30-105	
Terphenyl-d14 (S)	%	76	37-109	

LABORATORY CONTROL SAMPLE: T154793-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bis(2-chloroethyl)ether	ug/L	100.0	68.2	68	43-126	
2-Chlorophenol	ug/L	100.0	56.7	57	41-89	
Phenol	ug/L	100.0	16.0	16	17-120	BS06
1,4-Dichlorobenzene	ug/L	100.0	51.9	52	39-78	
Bis(2-chloroisopropyl)ether	ug/L	100.0	81.1	81	63-139	
N-Nitrosodi-n-propylamine	ug/L	100.0	75.4	75	14-198	
Nitrobenzene	ug/L	100.0	68.1	68	54-158	
Isophorone	ug/L	100.0	63.1	63	47-180	
2-Nitrophenol	ug/L	100.0	60.1	60	45-167	
2,4-Dimethylphenol	ug/L	100.0	57.8	58	1-173	
Bis(2-chloroethoxy)methane	ug/L	100.0	70.9	71	49-165	
1,2,4-Trichlorobenzene	ug/L	100.0	56.7	57	57-130	
2,4-Dichlorophenol	ug/L	100.0	58.6	59	54-122	
Naphthalene	ug/L	100.0	61.6	62	36-120	
4-Chloro-3-methylphenol	ug/L	100.0	56.4	56	45-93	
2,4,6-Trichlorophenol	ug/L	100.0	67.7	68	52-129	
2-Chloronaphthalene	ug/L	100.0	65.3	65	65-120	
Dimethyl phthalate	ug/L	100.0	67.8	68	1-120	
Acenaphthylene	ug/L	100.0	59.1	59	54-126	
2,6-Dinitrotoluene	ug/L	100.0	75.5	75	68-137	
Acenaphthene	ug/L	100.0	68.4	68	60-132	
2,4-Dinitrotoluene	ug/L	100.0	65.4	65	48-127	
4-Nitrophenol	ug/L	100.0	<20	12	13-129	BS06, J
2,4-Dinitrophenol	ug/L	100.0	66.3	66	1-173	
Diethyl phthalate	ug/L	100.0	69.2	69	1-120	
Fluorene	ug/L	100.0	70.2	70	70-120	
4-Chlorophenyl phenyl ether	ug/L	100.0	67.6	68	38-145	

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LABORATORY CONTROL SAMPLE: T154793-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
4,6-Dinitro-2-methylphenol	ug/L	100.0	67.3	67	53-130	
1,2-Diphenylhydrazine	ug/L	100.0	71.2	71	55-115	
4-Bromophenyl phenyl ether	ug/L	100.0	72.4	72	65-120	
Phenanthrene	ug/L	100.0	71.6	72	65-120	
Anthracene	ug/L	100.0	74.6	75	43-120	
Di-n-butyl phthalate	ug/L	100.0	70.9	71	8-120	
Fluoranthene	ug/L	100.0	79.1	79	43-121	
Pyrene	ug/L	100.0	78.2	78	70-120	
Butyl benzyl phthalate	ug/L	100.0	81.8	82	1-140	
Benzo (a) anthracene	ug/L	100.0	72.8	73	43-133	
Chrysene	ug/L	100.0	73.5	73	44-140	
Bis(2-ethylhexyl)phthalate	ug/L	100.0	76.3	76	29-137	
Di-n-octyl phthalate	ug/L	100.0	75.5	75	19-132	
Benzo (b) fluoranthene	ug/L	100.0	76.8	77	42-140	
Benzo (k) fluoranthene	ug/L	100.0	71.4	71	25-146	
Benzo (a) pyrene	ug/L	100.0	73.9	74	32-148	
Indeno (1,2,3-cd) pyrene	ug/L	100.0	74.5	74	1-151	
Dibenz (a,h) anthracene	ug/L	100.0	69.5	69	1-200	
Benzo (g,h,i) perylene	ug/L	100.0	82.4	82	1-195	
2-Fluorophenol (S)	%	100.0	26.6	27	20-53	
Phenol-d5 (S)	%	100.0	15.1	15	11-40	
Nitrobenzene-d5 (S)	%	100.0	62.8	63	36-103	
2-Fluorobiphenyl (S)	%	100.0	64.0	64	36-119	
2,4,6-Tribromophenol (S)	%	100.0	69.8	70	30-105	
Terphenyl-d14 (S)	%	100.0	77.7	78	37-109	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154928

Analysis Description: 625 Benzidines SIM

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1 SIM

METHOD BLANK: T154928-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	55	38-130	
2-Fluorobiphenyl (S)	%	81	39-118	
Terphenyl-d14 (S)	%	77	14-96	

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LABORATORY CONTROL SAMPLE: T154928-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.000	0.851	43	5.7-62	
3,3'-Dichlorobenzidine	ug/L	2.000	1.47	74	51-136	
Nitrobenzene-d5 (S)	%	1.000	0.600	60	38-130	
2-Fluorobiphenyl (S)	%	1.000	0.995	99	39-118	
Terphenyl-d14 (S)	%	1.000	0.927	93	14-96	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T154763

Analysis Description: TTO Volatiles

QC Batch Method: EPA 624 Purge and Trap

Analysis Method: EPA 624.1

METHOD BLANK: T154763-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chloromethane	ug/L	<1.0	1.0	
Vinyl chloride	ug/L	<1.0	1.0	
Bromomethane	ug/L	<1.0	1.0	
Chloroethane	ug/L	<2.0	2.0	
Acrolein	ug/L	<10	10	
1,1-Dichloroethene	ug/L	<1.0	1.0	
Methylene chloride	ug/L	<5.0	5.0	
Acrylonitrile	ug/L	<2.0	2.0	
trans-1,2-Dichloroethene	ug/L	<1.0	1.0	
1,1-Dichloroethane	ug/L	<1.0	1.0	
Chloroform	ug/L	<1.0	1.0	
1,1,1-Trichloroethane	ug/L	<1.0	1.0	
Carbon tetrachloride	ug/L	<1.0	1.0	
Benzene	ug/L	0.330	1.0	J
1,2-Dichloroethane	ug/L	<1.0	1.0	
Trichloroethene	ug/L	<1.0	1.0	
1,2-Dichloropropane	ug/L	<1.0	1.0	
cis-1,2-Dichloroethene	ug/L	<1.0	1.0	
Bromodichloromethane	ug/L	<1.0	1.0	
2-Chloroethylvinyl ether	ug/L	<5.0	5.0	
cis-1,3-Dichloropropene	ug/L	<1.0	1.0	
Toluene	ug/L	<1.0	1.0	
trans-1,3-Dichloropropene	ug/L	<1.0	1.0	
1,1,2-Trichloroethane	ug/L	<1.0	1.0	
Tetrachloroethene	ug/L	<1.0	1.0	
Dibromochloromethane	ug/L	<1.0	1.0	
Chlorobenzene	ug/L	<1.0	1.0	

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METHOD BLANK: T154763-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Ethylbenzene	ug/L	<1.0	1.0	
Bromoform	ug/L	<1.0	1.0	
1,1,2,2-Tetrachloroethane	ug/L	<1.0	1.0	
1,3-Dichlorobenzene	ug/L	<1.0	1.0	
1,4-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichloroethane-d4 (S)	%	93	68-133	
Toluene-d8 (S)	%	109	75-120	
4-Bromofluorobenzene (S)	%	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	105	72-127	

LABORATORY CONTROL SAMPLE: T154763-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chloromethane	ug/L	50.00	44.9	90	1-205	
Vinyl chloride	ug/L	50.00	46.4	93	5-195	
Bromomethane	ug/L	50.00	50.5	101	15-185	
Chloroethane	ug/L	50.00	50.3	101	40-160	
Acrolein	ug/L	50.00	29.4	59	60-140	BS06
1,1-Dichloroethene	ug/L	50.00	56.8	114	50-150	
Methylene chloride	ug/L	50.00	52.4	105	60-140	
Acrylonitrile	ug/L	50.00	25.2	50	60-140	BS06
trans-1,2-Dichloroethene	ug/L	50.00	53.7	107	70-130	
1,1-Dichloroethane	ug/L	50.00	48.9	98	70-130	
Chloroform	ug/L	50.00	50.6	101	70-135	
1,1,1-Trichloroethane	ug/L	50.00	50.5	101	70-130	
Carbon tetrachloride	ug/L	50.00	56.1	112	70-130	
Benzene	ug/L	50.00	53.2	106	65-135	
1,2-Dichloroethane	ug/L	50.00	42.4	85	70-130	
Trichloroethene	ug/L	50.00	53.0	106	65-135	
1,2-Dichloropropane	ug/L	50.00	50.9	102	35-165	
cis-1,2-Dichloroethene	ug/L	50.00	53.0	106	70-130	
Bromodichloromethane	ug/L	50.00	52.3	105	65-135	
2-Chloroethylvinyl ether	ug/L	50.00	44.3	89	1-225	
cis-1,3-Dichloropropene	ug/L	50.00	48.5	97	25-175	
Toluene	ug/L	50.00	60.1	120	70-130	
trans-1,3-Dichloropropene	ug/L	50.00	43.9	88	50-150	
1,1,2-Trichloroethane	ug/L	50.00	50.8	102	70-130	
Tetrachloroethene	ug/L	50.00	64.2	128	70-130	
Dibromochloromethane	ug/L	50.00	54.3	109	70-135	
Chlorobenzene	ug/L	50.00	55.6	111	65-135	

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LABORATORY CONTROL SAMPLE: T154763-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Ethylbenzene	ug/L	50.00	55.7	111	60-140	
Bromoform	ug/L	50.00	56.0	112	70-130	
1,1,2,2-Tetrachloroethane	ug/L	50.00	53.4	107	60-140	
1,3-Dichlorobenzene	ug/L	50.00	51.6	103	70-130	
1,4-Dichlorobenzene	ug/L	50.00	50.4	101	65-135	
1,2-Dichlorobenzene	ug/L	50.00	48.7	97	65-135	
1,2-Dichloroethane-d4 (S)	%	30.00	26.2	87	68-133	
Toluene-d8 (S)	%	30.00	33.1	110	75-120	
4-Bromofluorobenzene (S)	%	30.00	33.2	111	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.00	29.6	99	72-127	

Trace Project ID: 24H0470

Client Project ID: NPDES AMR.- August 2024

QC Batch: T155134

Analysis Description: Phenols, Total

QC Batch Method: EPA 420.4 Rev. 1.0

Analysis Method: EPA 420.4 Rev. 1.0

METHOD BLANK: T155134-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenolics	ug/L	<10	10	

LABORATORY CONTROL SAMPLE: T155134-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenolics	ug/L	100.0	96.0	96	73-110	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

Definitions and Notes

Trace ID: T154763-BS1

Analysis: EPA 624.1

Acrolein	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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Acrylonitrile	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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Trace ID: T154793-BS1

Analysis: EPA 625.1

4-Nitrophenol	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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Phenol	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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24H0470

Cadillac, City of

Project Manager: Tim Brewer

Sample Log In Checklist

Date: 08-08-24	Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-10 (CF: -1.0°C)	IR-12 (CF: 0.0°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.1°C)	Temp Blank	Client Sample
Time: 1640									
Initials: CM									
Package Description: cooler									
Package Temp °C	0.7	0.7							
Representative Sample Temp °C	1.5	1.3							

Sample Receipt

Yes	No	
☒	☐	Received on ice or other coolant
☒	☐	Ice still present upon receipt
☐	☒	Custody seals present
☒	☐	Trace Courier
☐	☐	Client Drop-off
☐	Yes	☐ No
☐	UPS	Custody seals intact (if applicable)
☐	Fed Ex	☐ US Mail
☐	Other	

Sample Condition

Yes	No	N/A	
☒	☐	☐	All sample containers arrived unbroken and labeled
☒	☐	☐	Sufficient sample to run requested analyses
☒	☐	☐	Correct chemical preservative added to samples
☒	☐	☐	Samples preserved at Trace
☒	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)
☐	☐	☐	☒ pH 0-2.5 (Lot: HC311850)
☐	☐	☐	☐ pH 11.0-13.0 (Lot: HC022540)
☐	☐	☐	☐ Other
☒	☐	☐	Air bubbles absent from VOAs

Chain of Custody (COC)

Yes	No	
☒	☐	All bottle labels agree with COC
☒	☐	COC filled out properly
☒	☐	COC signed by client

Notes:

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